

Physics-informed neural network for co-estimation of state of health, remaining useful life, and short-term degradation path in Lithium-ion batteries

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HIGHLIGHTS

- A gated multi-task framework is developed for SOH, RUL, and S-DP prediction.
- A simplified transformer model extracts both temporal and feature information.
- The partial derivatives of task outputs reveal hidden physical insights.

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ABSTRACT

Lithium-ion batteries gradually degrade over time due to various internal and external factors. This degradation introduces significant safety and reliability risks, highlighting the importance of battery health management as a critical area of research. However, a major challenge remains to develop a universal health management approach that accommodates various battery materials, operating environments, and diverse tasks. To address this, we present a novel multi-task health management approach that combines a multi-task processing framework with a physics-informed neural network. By leveraging the co-design of shared and task-specific parameters alongside physics-informed feature extraction, the approach efficiently integrates the tasks of state of health estimation, remaining useful life prediction, and short-term degradation path forecasting. Specifically, the method captures voltage and current data before and after the constant voltage charging phase, and employs an improved transformer model to extract temporal information. An adaptive weighting method is then applied to integrate the task losses effectively. Experimental results demonstrate that the mean absolute percentage error (MAPE) of state of health (SOH) estimation is 0.75 %, the normalized deviation of short-term degradation path (S-DP) prediction is approximately 0.01, and the mean absolute error (MAE) of remaining useful life (RUL) prediction is 104 cycles. Model comparative experiments, sensitivity analyses of the training sample size, and transfer learning demonstrate that the proposed framework not only substantially improves prediction accuracy but also showcases strong generalization capabilities and practical applicability. This provides a novel research perspective for advancing battery management technologies.

1. Introduction

Amid the global warming crisis, electrochemical energy has emerged as a rapidly developing medium for energy storage and utilization, owing to its portability, high integration, and recyclability [1,2]. Among these, lithium-ion batteries are extensively employed in applications such as electric vehicles [3,4], consumer electronics [5], and energy

storage systems [6,7], due to their favorable environmental compatibility, high energy density, and excellent efficiency. However, over time, as the service life of lithium-ion batteries increases, they encounter issues such as lithium plating, loss of active materials, structural changes, and a rise in undesirable side reactions. These changes manifest as macroscopic performance degradation, including capacity fade, increased resistance, and elevated temperature [8]. Furthermore,

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excessive charging and discharging will significantly accelerate the aging process of the battery and increase the probability of thermal runaway [9,10].

State of Health (SOH) can be evaluated based on the capacity degradation or the increase in internal resistance of a battery. However, most researchers typically define SOH as the ratio of the remaining capacity to the rated capacity of a lithium-ion battery [11]. This definition is also adopted in this study. When the SOH drops below 80 %, the safety risks associated with the battery become significant, potentially leading to hazardous incidents such as thermal runaway, leakage, or short circuits [12]. Researchers have extensively studied SOH estimation for both individual batteries and battery packs to mitigate safety risks caused by excessive aging [13]. Effective health management of lithium batteries has become essential, with key tasks including SOH estimation, degradation path (DP) prediction, and remaining useful life (RUL) prediction. SOH estimation reflects the current state of the battery, while the DP illustrates how the SOH deteriorates over repeated charge-discharge cycles, and RUL directly indicates the remaining time until the battery is no longer viable. These tasks are complementary but distinct. Accurate and reliable battery health management is critical for prolonging battery life, mitigating safety risks, and ensuring user safety [14]. The battery health management tasks discussed above are of great significance in energy systems. On the one hand, by enabling earlier and more accurate health status monitoring, systems can dynamically adjust operating strategies to avoid over-discharging, and thermal runaway, thereby effectively extending battery lifespan and improving energy utilization efficiency. On the other hand, RUL prediction allows for early fault warnings, reducing the risk of sudden safety hazards. Moreover, as a key energy buffering component in energy systems, the health status of batteries directly influences the effectiveness and economic performance of energy storage dispatch strategies. Therefore, accurate assessment of battery health plays a crucial role in enhancing the reliability and cost-efficiency of energy system operations.

Currently, research on lithium battery health management technologies can be categorized into model-based, data-driven, and hybrid approaches [15]. The model-based approach utilizes mathematical equations to describe the complex degradation mechanisms of batteries, including empirical formulas, equivalent circuit models, and electrochemical models. Empirical formulas, due to their simplicity, have been widely studied. These include exponential [16], linear [17], and polynomial forms [18], which are derived by analyzing the factors influencing the battery degradation process and extrapolating experimental data. The equivalent circuit model, an approximation based on electrical principles, is another widely used method, particularly in the context of battery thermal modeling [19]. For example, Wang et al. [20], Zhang et al. [21] and Li et al. [22] employed the equivalent circuit model for lithium battery degradation. By neglecting the detailed kinetic principles within the battery, this model uses resistances and capacitances to simulate the degradation process, simplifying the calculations. However, the parameters in both the empirical formula and the equivalent circuit model lack direct physical interpretation, which limits the generalization ability of these models. In contrast, electrochemical models are grounded in fundamental material and energy conservation equations, offering a clearer explanation of the degradation mechanisms [23]. The most common electrochemical model used for battery health management is the pseudo-two-dimensional (P2D) model [24], along with its simplified versions, such as the single-particle model [25]. These models are particularly useful for studying the impacts of phenomena such as lithium plating, solid electrolyte interface thickening, and active material loss on battery degradation and aging [26]. However, electrochemical modeling is highly complex due to the need to define numerous parameters and solve a system of partial differential Eqs. [27]. Moreover, all three model-based approaches often require the estimation of parameters that must be adjusted based on the specific battery materials and operating conditions. This necessitates the use of methods such as Kalman filtering, genetic algorithms, and least squares

estimation to identify and track these parameters in a closed-loop manner, which further adds to the model's complexity [28].

In recent years, with the rapid advancement of deep learning and artificial intelligence, data-driven methods have gained widespread application in lithium-ion battery health management [11]. These methods are often referred to as 'black-box' models, as they focus on approximating the real model through data, without considering the underlying principles, and rely on techniques such as gradient descent for continuous refinement. Data-driven approaches for lithium battery health management can be broadly categorized into two groups: those that use only current-time information and those that incorporate multiple historical time points, depending on the temporal scale of the training data. The former, which utilizes only current-time information, offers advantages such as simpler network structures and high computational efficiency. For example, Ye et al. estimated SOH [29] using multilayer perception, Zhang et al. predicted RUL [30] with support vector machines, and Alipour et al. estimated RUL [31] using Gaussian process regression. Sun et al. utilized the bidirectional gated recurrent unit Attention model to estimate SOH [32]. However, these approaches are limited in their ability to capture both long-term and short-term temporal dependencies within the data, which may lead to reduced accuracy and hinder effective prediction of the degradation path. In contrast, by incorporating information from multiple historical time steps, time-series data can be leveraged more comprehensively, enabling iterative or sequence-to-sequence predictions of degradation paths. Many researchers have explored the use of deep learning models such as CNN [33,34], LSTM [35], and GRU [36] for battery health management. Although LSTM and GRU are designed to address challenges such as vanishing and exploding gradients in traditional RNNs and are effective for sequential modeling, they may still encounter difficulties in capturing very long-term dependencies or complex degradation dynamics. To address these challenges, models such as Bi-LSTM [37] (Bidirectional Long Short-Term Memory Network) and the Transformer series [38–40] have been introduced to better capture long-range dependencies. For example, despite their advantages, data-driven methods still face challenges, including poor performance when data is scarce, limited transferability across different domains, and insufficient generalization capabilities when exposed to varied datasets.

The physics-informed neural network is a hybrid approach that integrates physical knowledge into the model, addressing the challenges of poor interpretability and limited generalization capability commonly associated with data-driven methods. Deng et al. categorized physics-informed neural network into three types based on the location of physical information fusion: (1) using physical knowledge to guide the construction of the input space, (2) guiding the network structure, and (3) constraining the objective function [41]. Xu et al. applied the P2D electrochemical model along with measured voltage data to extract features for predicting the lithium battery degradation path [42]. Cui et al. utilized the P2D model to generate battery data and then applied a transformer model to correct errors, achieving ultra-early performance prediction based solely on charging and discharging strategies [43]. E et al. developed a multi-factor aging model that incorporates knee point dynamics and embeds electrochemical constraints into the loss function, enabling the joint prediction of the knee point and RUL [44]. Zhang et al. identified battery degradation using an equivalent circuit model based on voltage relaxation data, using the resistance and capacitance values from the model as network inputs to predict battery life [21]. Ma et al. extracted parameters from the IC curve using empirical formulas and employed them as network inputs to predict the RUL of the battery [45]. Ye et al. leveraged the monotonic relationship between the peak value of the IC curve and SOH to define the objective function for a physically constrained network, enhancing prediction accuracy [29]. Current research is primarily focused on using physical information to guide input space construction and applying simple data relationships to constrain the network's objective function. However, given the complexity of battery degradation models, using electrochemical partial

The paper is structured as follows: [Section 2](#) provides a detailed description of the dataset, including the preprocessing and feature extraction procedures. [Section 3](#) outlines the model architecture and the methodologies utilized. [Section 4](#) presents and analyzes the experimental results, while [Section 5](#) concludes the study.

The performance of lithium-ion batteries is influenced by both their structure and materials. The most commonly used battery types on the market are lithium iron phosphate (LFP) and ternary lithium batteries. Variations in materials, form factors, and production batches can lead to differences in capacity, rated voltage, and degradation characteristics across different batteries. Even for the same battery, factors such as operating environment (e.g., temperature) and charging-discharging strategies can result in divergent health state evolution patterns. Therefore, the development of a universal battery health management model that accommodates various materials and usage conditions is of paramount importance.

As shown in Fig. 1(a), we aim to develop a universal battery health management model by utilizing battery data under varying chemical compositions, operating temperatures, and charging-discharging strategies. The two datasets used in this study are derived from the works of Ma et al. [34] and Zhu et al. [46], referred to as the HUST dataset [34] and the TJU dataset [46], respectively. A total of 196,200 cycles from 198 batteries, all of which are 18650-type cells, are utilized. The basic parameters of both datasets are summarized in Table 1, while the cycle life curves for all batteries are illustrated in Fig. 1(b). The HUST dataset comprises 77 lithium iron phosphate (LFP)/graphite batteries with a rated capacity of 1.1 Ah and cut-off voltages ranging from 2.0 to 3.6 V. This dataset employs diverse multi-stage constant-current discharge strategies. The charging protocol is standardized: a 5C constant current charge to 80 % state of charge (SOC), followed by a 1C constant current charge to a cut-off voltage of 3.6 V, and finally a constant voltage charge until the current drops to 1/20C. All batteries are cycled continuously at 30 °C, with cycle lifetimes ranging from 1100 to 2700 cycles. This dataset allows for simulations of battery usage under varying discharge conditions. The TJU dataset consists of two types of ternary lithium batteries: nickel-cobalt-aluminum oxide (NCA)/graphite and nickel-cobalt-manganese oxide (NCM)/graphite, both with a rated capacity of 3.5 Ah. These batteries are tested at three different temperatures: 25 °C, 35 °C, and 45 °C, with each battery being subjected to testing at these specific temperatures individually. The charging protocol involves a constant current (0.25–1C) charge to a cut-off voltage of 4.2 V, followed by a constant voltage charge until the current decreases to 1/20C. The discharge strategy is relatively simple, using a 1C constant current rate to the cut-off voltage. Batteries are cycled until their capacity degrades to 71 % of the rated capacity, with cycle lifetimes ranging from 50 to 1250 cycles. This battery dataset covers a range of operating temperatures and reflects diverse user scenarios with significant variations in battery lifespan and degradation paths.



Table 1
Information of the battery data set.

Dataset	Cell type	Nominal capacity (Ah)	Cut-off voltage (V)	Experiment temperature (°C)	Number of cells
HUST	LiFePO ₄ (LFP)	1.1	2.0–3.6	30	77
TJU	Li _{0.86} Ni _{0.86} Co _{0.11} Al _{0.03} O ₂ (NCA)	3.5	2.65–4.2	25,35,45	66
	Li _{0.84} Ni _{0.83} Co _{0.11} Mn _{0.07} O ₂ (NCM)	3.5	2.5–4.2	25,35,45	55

2.2. Feature extraction and analysis

Extracting features that are strongly correlated with outcomes from generalized data is a key focus of research efforts [47]. In existing studies, numerous researchers have extracted features related to battery health from curves such as dQ/dV , relaxation voltage, temperature, resistance, incremental capacity (IC), and capacity degradation. However, these curves are often difficult to measure during actual battery operation due to equipment limitations and variable operating conditions. Consequently, we focus on extracting features from more universally available and easily measurable battery voltage and current curves. In practical applications, the discharge curves of batteries are often highly variable due to user demand, complicating feature extraction. In contrast, the charging process typically follows well-defined strategies, providing greater consistency and universality. A typical battery charging process involves multi-stage strategies: high-rate currents are used when the battery capacity is low, the charging rate is reduced as the capacity reaches a certain threshold, and finally, constant voltage charging is employed once the battery voltage reaches the cut-off value. This method is widely adopted for its efficient charging performance and effective battery protection. As shown in Fig. 1(c), inspired by Wang et al. [48], we extract a specific segment of the current

and voltage curves surrounding the constant voltage charging phase—prior to the battery reaching full charge—for feature extraction.

As shown in Fig. 1(d–e), current and voltage curve segments are extracted from all cycles of a battery. From these segments, we derive 20 statistical features for each cycle, including cycle number, accumulated charge, mean value, standard deviation, duration, kurtosis, skewness, curve slope, and curve entropy. The main purpose of introducing the current cycle number is to provide temporal context that reflects the battery's degradation stage, enabling the model to learn the evolution of health states over time [30,34,48,49]. While complete cycles are rare under real-world conditions, the equivalent full cycle metric, widely used in aging analysis, estimates usage by normalizing cumulative Ah throughput. Additionally, inspired by Severson et al. [49], to further capture cross-time-dimensional information, we compute the difference between each cycle's curve and the 10th cycle's curve, from which we extract the mean and standard deviation as additional features. These statistical features have been widely validated for battery degradation modeling due to their robustness and interpretability. We also introduced interval-based features from voltage and current curves for better efficiency and deployment. Furthermore, we introduced multi-scale feature fusion and physics-informed interaction modules into the model architecture. These designs enable the traditional features to

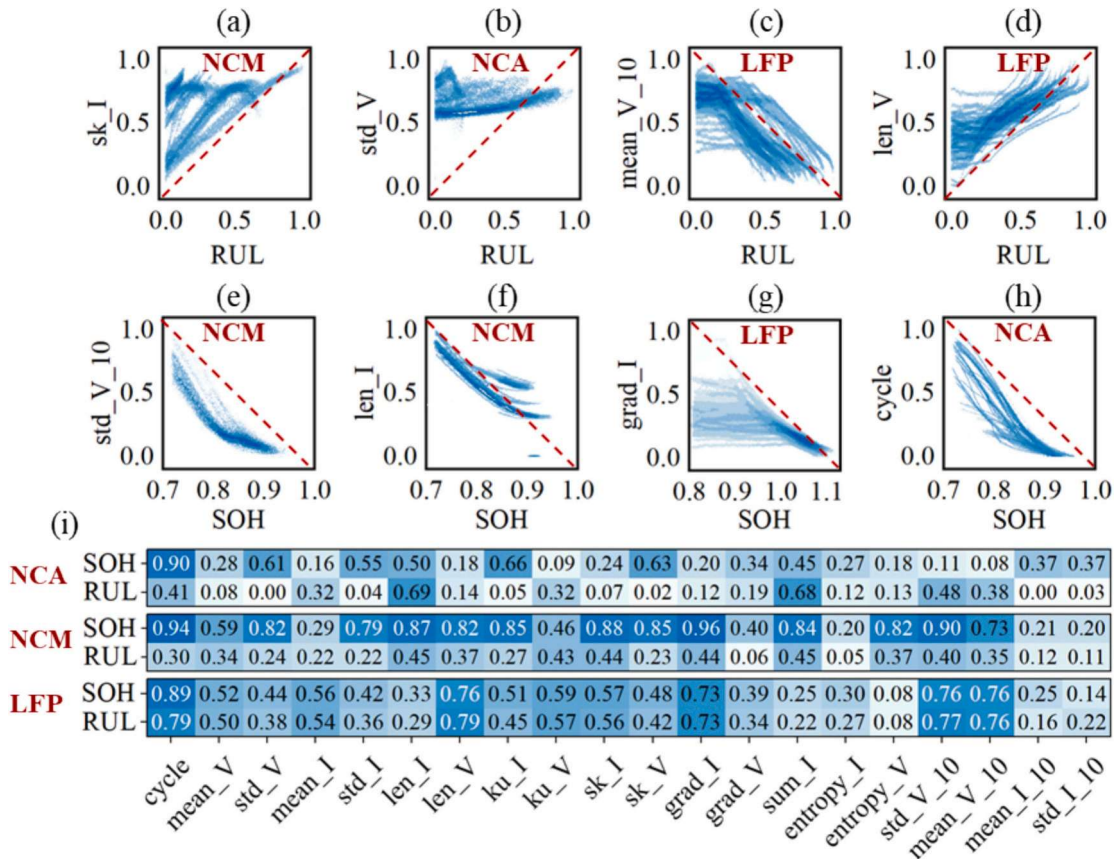


Fig. 2. Linear correlation between features and targets: (a–h) Diagonal plots; (i) Pearson correlation coefficients computed for features such as cycle number (cycle), mean value (mean), standard deviation (std), duration time (len), kurtosis (ku), skewness (sk), curve slope (grad), curve entropy (entropy), accumulated charge (sum), and cross-time metrics (e.g., mean voltage deviation from the 10th cycle, mean_V_10).

better capture inter-feature relationships, thereby improving the overall modeling performance. We believe that this joint design of features and model architecture effectively extends the expressive capacity of conventional features. The temperature variation and charge/discharge strategy are not used as input features, but are implicitly embedded in the data distribution and reflected in input signals such as voltage, current, and SOH [34,46,48]. Their impact is captured during the model training and evaluation process. In the future, we will explore the integration of innovative feature extraction techniques or new types of features, such as employing CNNs for automatic feature extraction, to enhance the expressiveness of learned representations.

As illustrated in Fig. 2, we analyze the Pearson correlation coefficients between these features and the SOH or RUL for batteries with different chemical compositions. Fig. 2(a-h) selectively demonstrates how certain features, after min-max normalization, vary with SOH or RUL. Darker blue areas in these figures indicate higher data density and data points closer to the diagonal suggest a stronger linear correlation between the feature and the corresponding target variable. From Fig. 2(i), it is evident that the chemical composition of the battery plays a crucial role in determining its properties, as the correlation between features and target values differs significantly across various compositions. The correlation between SOH and the features is stronger than that between RUL and the features, which is related to the time scales and task difficulties: SOH estimation is a short-term task, whereas RUL prediction is inherently a long-term task. The correlation coefficients for RUL in NCA and NCM batteries are relatively low, which can be explained by the dispersion of the battery data and the diversity of degradation patterns.

3. Methodology

We have developed a multi-task health management model for lithium-ion batteries, integrating a Customized Gate Control Model and

a Physics-Informed Neural Network (PINN). As illustrated in Fig. 3, we use 20 features extracted from the voltage and current curves surrounding the constant voltage charging phase of the battery as the model input. Here, t represents the current cycle number, and x denotes the statistical features extracted from the voltage and current data, as described in Section 2. The input is initially processed by shared expert models and task-specific expert models, then weighted and summed through a control gate, and finally fed into the tower models for different tasks. Following the initial tower outputs, the PINN is incorporated as a regularization term to enhance the model's performance. As the SOH estimation and degraded path predictor provide SOH at different time scales, and the latter yields large data volumes, adding it to the PINN would raise computational costs. Hence, we only use SOH and RUL to compute partial derivatives as the PINN input.

The expert models are based on MLP, while the SOH estimation, RUL prediction, and PINN utilize a simplified transformer model adapted for time-series data. We used Bi-LSTM for degradation path prediction because the task is best handled by coordinated encoder-decoder structures. To simplify the model, we used a unified architecture, avoiding the complexity of full transformer setups. Moreover, Bi-LSTM showed better temporal modeling in our experiments, while the transformer's strengths are in parallelism and feature-temporal fusion. The number of shared experts, task-specific experts, and the number of model layers are all hyperparameters that require tuning. We train the model separately on battery datasets with different electrode materials to demonstrate its generalization capability. Each dataset typically includes various charging/discharging strategies and environmental temperatures. This approach aims to create a multi-task lithium-ion battery health management model capable of deeply extracting physical information from the data. The goal of the proposed framework is not merely to combine multiple tasks, but to exploit their inherent interdependencies through collaborative modeling. This enables the model to better capture the fundamental degradation mechanisms,

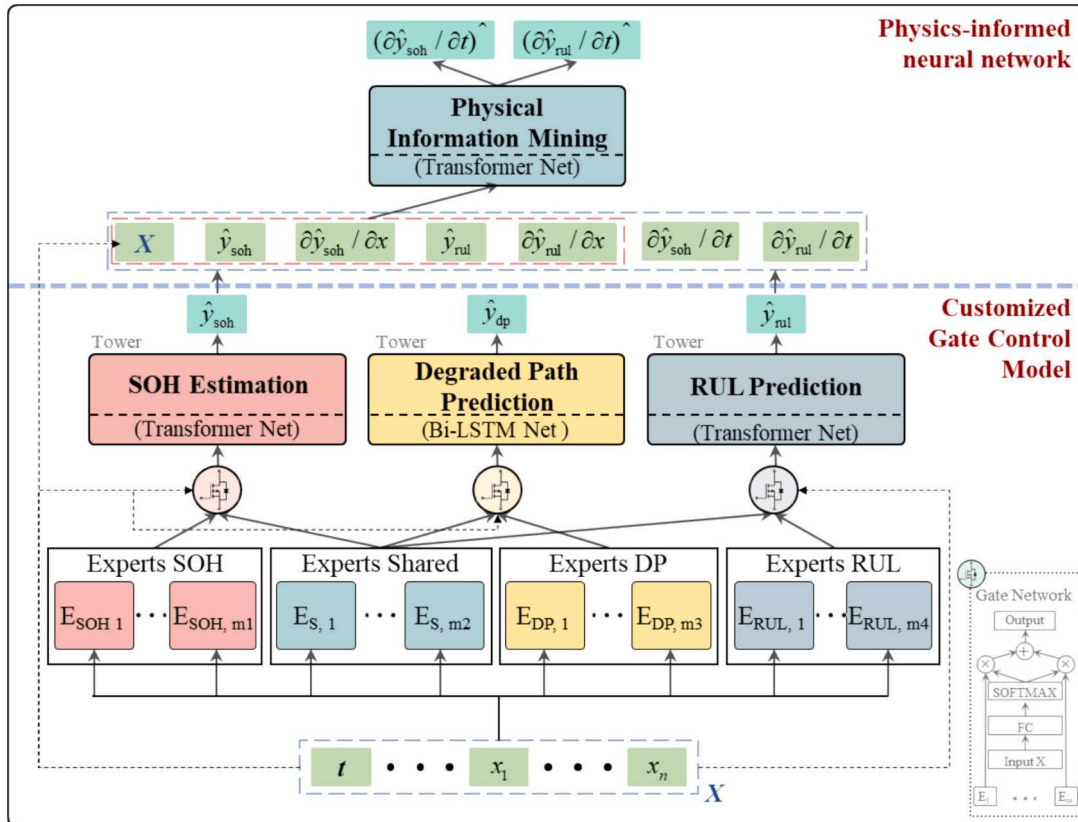


Fig. 3. The framework schematic diagram of the model.

thereby enhancing both prediction accuracy and physical consistency.

3.1. Customized gate control model

The model proposed in this paper needs to simultaneously handle three tasks: SOH estimation, S-DP prediction, and RUL prediction. Among these, S-DP prediction and SOH estimation are battery capacity estimations at different time scales. Utilizing a single model to manage multiple tasks under identical input conditions has been shown to offer benefits such as reduced learning parameters and increased efficiency, as different tasks can share network parameters and information [50,51]. However, this approach may also lead to negative transfer and the seesaw phenomenon [52]. Negative transfer occurs when the correlation between tasks is weak or even contradictory, rendering shared information ineffective or potentially harmful, thereby impairing performance across all tasks. The seesaw phenomenon arises when tasks compete for resources or exhibit conflicting objectives, causing improvements in one task to come at the expense of others. To mitigate these issues, we introduce the Customized Gate Control (CGC) model, which integrates both shared and task-specific layers, using a customized gating mechanism to selectively weigh and combine the outputs of these layers [50,52]. This approach offers an enhancement over traditional multi-task models that rely solely on either shared or task-specific layers.

The CGC model consists of three main components: multiple expert modules at the bottom, a gating module in the middle, and tower modules at the top. Each expert module is composed of multiple expert networks, and the number of these networks is a hyperparameter. Taking the SOH estimation task as an example, the output of the expert module is as follows:

$$S_{\text{soh}}(\mathbf{X}) = \left[\mathbf{E}_{(\text{SOH},1)}^T, \dots, \mathbf{E}_{(\text{SOH},m_1)}^T, \mathbf{E}_{(\text{S},1)}^T, \dots, \mathbf{E}_{(\text{S},m_2)}^T \right]^T \quad (1)$$

where $\mathbf{E}_{\text{SOH}}$ represents the output of the SOH estimation expert network, and $\mathbf{E}_{\text{S}}$ represents the output of the shared expert network. The output of the expert module is transmitted to the tower layer network of each task after being weighted by the gating network. The gated network uses feature $\mathbf{X}$ as input, generates weights after passing through a single-layer fully connected network, and uses weights to sum the output of the expert module. The calculation process is as follows:

$$w_{\text{soh}}(\mathbf{X}) = \text{softmax}(\mathbf{W}_g^T \mathbf{X} + \mathbf{b}_g) \quad (2)$$

$$g_{\text{soh}}(\mathbf{X}) = w_{\text{soh}}(\mathbf{X}) S_{\text{soh}}(\mathbf{X}) \quad (3)$$

After obtaining the output of the gating module, the result of the SOH estimation task can be expressed as follows:

$$y_{\text{soh}}(\mathbf{X}) = t_{\text{soh}}(g_{\text{soh}}(\mathbf{X})) \quad (4)$$

where t_{soh} represents the tower network of the SOH estimation task.

In our implementation, we use four expert modules, as this configuration offers a good trade-off between performance and computational efficiency. The shared expert module includes two fully connected neural networks, while each of the remaining three expert modules contains a single fully connected neural network. All expert networks have two hidden layers with 25 neurons each, and use ReLU as the activation function. Each expert network takes 20-dimensional input features and outputs an 8-dimensional learned feature vector, which is subsequently integrated by the gate controller.

3.2. Simplified time series transformer model

Data collected from each charging cycle forms a time-series dataset, and fully leveraging the characteristics of these time series is essential for enhancing the accuracy of battery health management models. The

Transformer model, with its multi-head attention mechanism, is particularly effective at extracting relevant information from time-series data. Its parallel computation capabilities enable it to capture both short- and long-range dependencies, thereby improving model accuracy and computational efficiency [53]. The Transformer model addresses the limitations of traditional time-series models, such as the vanishing and exploding gradients encountered by recurrent neural networks (RNNs) when processing long sequences, as well as the excessive parameter count and computational inefficiency associated with LSTM networks. Originally developed for machine translation, the Transformer's applicability has expanded, with notable success in fields such as image processing [54], speech recognition [55], and time series forecasting [56]. Today, many state-of-the-art AI models, including GPT and BERT, are built upon the Transformer framework, underscoring its versatility and robustness in handling multimodal tasks.

Many scholars have improved the Transformer model to make it more suitable for time series forecasting tasks [39,50,57]. The model proposed in this paper focuses on enhancing the Transformer encoder, incorporating modifications to the input encoding and output mapping layers to better accommodate time series prediction tasks. The decoder structure, teacher forcing mechanism, and the difference in input logic between training and inference are not included in the model. The adopted model structure is illustrated in Fig. 4. It consists of six components: position encoding, multi-head attention computation, residual network, layer normalization, feedforward neural network, and mapping network. Since the self-attention mechanism does not have any sequential dependencies, the input features $\mathbf{X}$ must first be encoded with position information, incorporating both the temporal and feature dimensions. A common position encoding scheme uses sine and cosine functions, and the computation process can be expressed by the following formula [53]:

$$PE(X) = \begin{cases} X + \sin(t/10000^{2p/m}), p = 2k \\ X + \cos(t/10000^{2p/m}), p = 2k + 1 \end{cases} \quad (5)$$

Here, t represents the position in the time dimension, p represents the position in the feature dimension, and m denotes the maximum number of features. After the position encoding, the data enters the multi-head attention mechanism layer. This layer utilizes multiple scaled dot-product self-attention mechanisms to capture the complex correlations of data across both the time and feature dimensions. The input data is multiplied by weight matrices to obtain the query matrix $\mathbf{Q}$, the key matrix $\mathbf{K}$, and the value matrix $\mathbf{V}$, which are then used to calculate the attention scores. The specific calculation formula is as follows:

$$\mathbf{Q}_i = \mathbf{X}\mathbf{W}_i^q, \mathbf{K}_i = \mathbf{X}\mathbf{W}_i^k, \mathbf{V}_i = \mathbf{X}\mathbf{W}_i^v \quad (6)$$

$$\mathbf{A}_i = \text{softmax}\left(\frac{\mathbf{Q}_i(\mathbf{K}_i)^T}{\sqrt{d_k}}\right)\mathbf{V}_i \quad (7)$$

where $\mathbf{Q}_i$ the query matrix of the i -th dot product attention calculation, $\mathbf{A}_i$ represents the output of the i -th dot product attention calculation, d_k represents the dimension of the $\mathbf{K}$, $\mathbf{W}$ is the weight matrix that can be learned. To focus on the information of features in different subspaces, the attention computer mechanism of multiple heads is introduced. The final output of the multi-head attention layer can be expressed by the following formula:

$$MHA(\mathbf{X}) = \text{concat}(\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_i, \dots, \mathbf{A}_h) \mathbf{W}^h \quad (8)$$

where concat represents the merging of matrices.

As shown in Fig. 4 (a), both the multi-head attention computation layer and the feedforward network layer have a skip connection, allowing the input to bypass these layers and be directly added to the output. This is based on the residual network proposed by He et al. [58], which helps to avoid the vanishing gradient problem and facilitates the

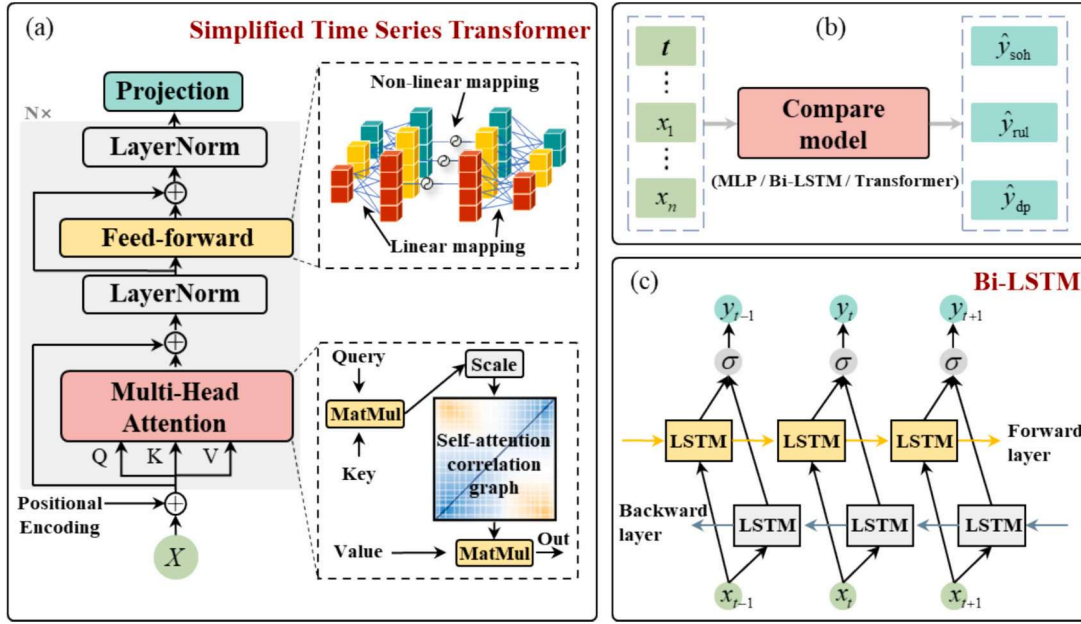


Fig. 4. (a) The Improved Transformer Model, (b) Comparison Model, and (c) Bi-LSTM Model.

construction of deeper networks. Layer normalization normalizes the different features within the same time sample. Unlike batch normalization, layer normalization normalizes across the feature dimension rather than the batch dimension. Both are used to stabilize numerical variations, thereby enhancing the robustness of the model. The feed-forward neural network consists of two linear layers and an activation function. The data is first linearly mapped to a higher-dimensional space, then subjected to nonlinear computation, and finally mapped back to a lower-dimensional space. This process enhances the model's nonlinear mapping capabilities and further uncovers the relationships between features. The mapping layer uses an MLP to convert the data into the final output. It is the most basic type of neural network, and its single-layer computation process can be expressed by the following formula.

$$h_i = \text{ReLU}(\mathbf{W}_i^T h_{i-1} + \mathbf{b}_i) \quad (9)$$

where $\mathbf{W}$ represents the weight parameter, h_i and h_{i-1} represents the output and input of this layer of network, respectively. $\mathbf{b}_i$ representing bias. ReLu is a commonly used activation function in deep learning. Its function is to change the negative value of the input into zero, and the positive value remains unchanged.

3.3. Physics-informed neural network

In recent years, PINN has been widely applied in fields such as fluid thermal simulation and biomedical research. Positioned between data-driven models and physics-based models, PINN combines the advantages of both approaches [59]. By directly incorporating physical laws (such as partial differential equations, conservation laws, or constraints) into the loss function, PINN not only effectively leverages available observational data but also adheres to known physical principles, thereby enhancing the model's generalization ability and accuracy. However, for domains with limited measurement data or complex physical mechanisms, the traditional approach of explicitly integrating physical laws into the model is not feasible. The lithium battery health management primarily involves constructing a capacity degradation model. The factors influencing battery degradation are complex and varied, including temperature, loss of active materials, charging/discharging strategies, and the battery's material composition, among others. To describe the degradation process of lithium batteries,

researchers have proposed numerous empirical formulas, such as exponential models, linear models, and polynomial models. However, these models are overly simplistic and mostly use time as the sole variable, making it difficult to accurately capture the complex degradation process of lithium batteries. As a result, many scholars have focused on directly using the partial derivatives of the model output data to construct implicit physical laws, integrating them into the loss function to further enhance the exploitation of physical principles [59–62].

Therefore, we construct a deep hidden physics model by calculating the partial derivatives of the output from the multi-task model and integrating it with the data-driven approach to build a PINN. To maintain generality, we assume that the capacity model of the lithium battery can be expressed by the following formula:

$$\mathbf{y} = f(\mathbf{t}, \mathbf{x}) \quad (10)$$

Here, t represents time, and $\mathbf{x}$ represents factors influencing degradation, such as temperature and charging/discharging strategies. However, these factors cannot be directly measured, so we represent $\mathbf{x}$ using the features extracted in this paper. Furthermore, the degradation model can be derived by taking the partial derivative of the capacity model for time:

$$\frac{\partial \mathbf{y}}{\partial t} = u(\mathbf{t}, \mathbf{x}, \mathbf{y}; \theta) \quad (11)$$

Here, u represents the degradation model, and the parameters θ are used to explicitly express the degradation model. However, since the explicit form of the degradation model is unknown, we can only approximate it as follows:

$$\mathbf{y}_t \approx u(\mathbf{t}, \mathbf{x}, \mathbf{y}, \mathbf{y}_x, \mathbf{y}_{xx}, \dots; \theta) \quad (12)$$

where $\mathbf{y}_x$ represents the first-order partial derivative of $\mathbf{y}$ for $\mathbf{x}$. Furthermore, we can incorporate an equation of the following form into the data-driven model:

$$\text{DeepHPM} : \mathbf{y}_t - u(\mathbf{t}, \mathbf{x}, \mathbf{y}, \mathbf{y}_x, \mathbf{y}_{xx}, \dots; \theta) = 0 \quad (13)$$

By using neural networks, we can learn Eq. (13) and enable the model to satisfy this hidden physical equation. Through automatic differentiation, we compute the partial derivatives of the target estimates from the data-driven model output and then use these as inputs to the network that learns the hidden physical laws. This approach strengthens

the model's ability to extract physical information from the data, enhancing both its generalization capability and interpretability. We extend the input space by incorporating the partial derivatives of both SOH and RUL outputs with respect to the input features. Importantly, we deliberately exclude temporal derivatives of the target outputs to prevent information leakage. The PINN receives only the input features, the task outputs, and their spatial (feature-wise) partial derivatives. This design choice, which emphasizes physical consistency and mitigates information leakage, has also been adopted in studies [61–63].

3.4. Weight coefficients tuning and performance indicators

The loss function of the multi-task learning model consists of multiple components, and determining the loss weights between the different objective functions is crucial for the model's performance [64]. Manually setting the weight values makes the hyperparameter tuning process cumbersome and time-consuming, limiting the model's generalization ability. Therefore, we introduce automatic hyperparameter tuning techniques, allowing the weights to adaptively change during the training process. Common adaptive tuning methods include Gradient Normalization [65], Dynamic Weight Averaging [66], and Uncertainty Weighting [64], each of which adjusts the weight based on the magnitude or rate of change of the task losses. Considering the model in this paper, we adopt the Uncertainty Weighting method to calculate the loss weights. This method is built on the homoscedastic uncertainty principle of Bayesian models and calculates the loss weights by measuring the uncertainty exhibited by the model's tasks without relying on input data [64]. Assuming that the likelihood of the task target data follows a Gaussian distribution, the mean is defined by the model's output, as shown in the following equation:

$$p(\mathbf{y}|\mathbf{f}(\mathbf{x})) = \mathcal{N}(\mathbf{f}(\mathbf{x}), \sigma^2) \quad (14)$$

where $\mathbf{y}$ represents the output of the task, and $\mathbf{x}$ represents the input of the task. σ is the observation noise of the model data, which is a learnable parameter during computation and will be updated via back-propagation through gradient. A similar setup is required for each task. The log-likelihood can be expressed in the following form:

$$\log p(\mathbf{y}|\mathbf{f}(\mathbf{x})) \propto -\frac{1}{2\sigma^2} \|\mathbf{y} - \mathbf{f}(\mathbf{x})\|^2 - \log \sigma \quad (15)$$

In the target outputs of this paper, there are four losses. The derivation can be expressed in the following form:

$$\begin{aligned} & -\log p(\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3, \mathbf{y}_4 | \mathbf{f}_1(\mathbf{x}), \mathbf{f}_2(\mathbf{x}), \mathbf{f}_3(\mathbf{x}), \mathbf{f}_4(\mathbf{x})) \\ &= -\log [p(\mathbf{y}_1 | \mathbf{f}_1(\mathbf{x})) \cdot p(\mathbf{y}_2 | \mathbf{f}_2(\mathbf{x})) \cdot p(\mathbf{y}_3 | \mathbf{f}_3(\mathbf{x})) \cdot p(\mathbf{y}_4 | \mathbf{f}_4(\mathbf{x}))] \\ &= -\log p(\mathbf{y}_1 | \mathbf{f}_1(\mathbf{x})) - \log p(\mathbf{y}_2 | \mathbf{f}_2(\mathbf{x})) - \log p(\mathbf{y}_3 | \mathbf{f}_3(\mathbf{x})) - \log p(\mathbf{y}_4 | \mathbf{f}_4(\mathbf{x})) \\ &= \mathcal{L}_1/2\sigma_1^2 + \mathcal{L}_2/2\sigma_2^2 + \mathcal{L}_3/2\sigma_3^2 + \mathcal{L}_4/2\sigma_4^2 + \log \sigma_1 \sigma_2 \sigma_3 \sigma_4 \end{aligned} \quad (16)$$

where $\mathcal{L}$ represents the loss function of the task, and the training process is to minimize Eq. (16). In this paper, the mean squared error is used as the loss function during the model training process, and it is expressed as follows:

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (\mathbf{y}_i - \hat{\mathbf{y}}_i)^2 \quad (17)$$

where $\mathbf{y}_i$ represents the model's target, and $\hat{\mathbf{y}}_i$ represents the model's estimated output. The model's loss function is then expressed as follows:

$$\mathcal{L}_{\text{soh}} = \frac{1}{N} \sum_{i=1}^N |\mathbf{y}_{\text{soh}} - \hat{\mathbf{y}}_{\text{soh}}|^2 \quad (18)$$

$$\mathcal{L}_{\text{rul}} = \frac{1}{N} \sum_{i=1}^N |\mathbf{y}_{\text{rul}} - \hat{\mathbf{y}}_{\text{rul}}|^2 \quad (19)$$

$$\mathcal{L}_{\text{dp}} = \frac{1}{N} \sum_{i=1}^N |\mathbf{y}_{\text{dp}} - \hat{\mathbf{y}}_{\text{dp}}|^2 \quad (20)$$

$$\mathcal{L}_{\text{physics}} = \frac{1}{N} \sum_{i=1}^N \left| \left(\frac{\partial \hat{\mathbf{y}}_{\text{soh}}}{\partial \mathbf{t}} \right) - \left(\frac{\partial \mathbf{y}_{\text{soh}}}{\partial \mathbf{t}} \right) \right|^2 + \frac{1}{N} \sum_{i=1}^N \left| \left(\frac{\partial \hat{\mathbf{y}}_{\text{rul}}}{\partial \mathbf{t}} \right) - \left(\frac{\partial \mathbf{y}_{\text{rul}}}{\partial \mathbf{t}} \right) \right|^2 \quad (21)$$

$$\mathcal{L} = \alpha \mathcal{L}_{\text{soh}} + \beta \mathcal{L}_{\text{dp}} + \chi \mathcal{L}_{\text{rul}} + \varepsilon \mathcal{L}_{\text{physics}} \quad (22)$$

Additionally, we evaluate the model's performance using the root mean square error (RMSE) and mean absolute percentage error (MAPE). Their calculation formulas are as follows:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\mathbf{y}_i - \hat{\mathbf{y}}_i)^2} \quad (23)$$

$$\text{MAPE} = \frac{1}{N} \sum_{i=1}^N \left| \frac{\mathbf{y}_i - \hat{\mathbf{y}}_i}{\mathbf{y}_i} \right| \times 100\% \quad (24)$$

3.5. Hyperparameter selection

The proposed model is optimized using an end-to-end joint training approach, in which all modules are trained collectively through back-propagation after the overall architecture is constructed. The network involves hyperparameters that require configuration, and their settings influence the performance of the model. Table 2 summarizes the hyperparameters of the proposed framework, including details of the network architecture (e.g., number of layers, input/output dimensions for each module) and training configuration. These hyperparameters were manually tuned based on extensive prior experience and commonly adopted practices in relevant literature. The Adam optimizer was consistently used during training. The model was trained separately for batteries with different chemistries, each covering a range of ambient temperatures and charge/discharge protocols.

4. Results and discussion

After extracting features from the charging current and voltage data, the dataset is divided into training, validation, and test sets, following an approximate ratio of 70 %:15 %:15 %. To ensure fairness, the test and validation sets are evenly sampled from different experimental measurement batches. The data is preprocessed using min-max normalization, with the min and max values derived from the training set applied to standardize both the validation and test sets, thereby preventing data leakage. The input consists of historical data from 30 charging cycles. The model's performance is evaluated through a detailed analysis of the results across different tasks and datasets, with comparisons made to existing models. Additionally, sensitivity analyses of the training sample size and transfer learning experiments are conducted to assess the robustness and adaptability of the proposed model.

4.1. SOH estimation

The SOH of lithium-ion batteries is a critical indicator for evaluating battery aging. Accurate SOH estimation not only provides users with reliable data on the battery's maximum usable capacity, which facilitates the adjustment of charging strategies and optimization of usage, but also serves as a valuable reference for developers in areas such as fast-charging strategy design and battery material composition optimization. State estimation was performed using the proposed model based on 20 features extracted from data segments around the constant voltage charging phase. The results are shown in Fig. 5(b), (f), and (j), where the x-axis represents the true data labels, and the y-axis represents the model's predicted results. The closer the data points are to the diagonal

Table 2
Hyperparameter selection.

Module	Parameter	Value	Parameter	Value
CGC	Number of experts in CGC (SOH / Shared / DP / RUL)	1/2/1/1	Fully connected layers	2
	Input / Output dimensions	20/8	Neurons in layer	25
Gate network	Input / Output dimensions	44/8		
	Number of encoder layers	3	Feed-forward layer width	50
Tower for SOH and RUL	Number of attention heads	2	Input / Output dimensions	8/1
	Number of layers	1	Neurons in layer	20
Tower for DP	Encoder / Decoder network type	Bi-LSTM		
	Number of encoder layers	3	Feed-forward layer width	50
PINN	Number of layers	2		
	Batch size	256	Optimizer	Adam
Training configuration	Maximum training epochs	1000	Regularization strategies	Dropout
	Learning rates	0.01	Scheduling strategy	Cosine annealing

line, the better the alignment between the predicted results and the true values. To validate the efficacy of the proposed model, we compared its prediction performance with that of MLP, Bi-LSTM, and the simplified Transformer model. The detailed comparison results are presented in Table 3, and the schematic of the comparison models is shown in Fig. 4 (b). To ensure experimental fairness, all models were configured with identical parameter scales, optimizers, loss functions, and consistent learning rate settings. Additionally, to mitigate the effects of randomness in the network computation process, 10 independent runs were performed for each experiment, with the results averaged across these runs. From the experimental outcomes on three distinct material datasets, the proposed model demonstrated superior performance in SOH estimation. Specifically, for the LFP, NCA, and NCM battery datasets, the MAPE is 0.85 %, 1.08 %, and 0.75 %, respectively. In comparison, Wang et al. [48] reported MAPE values of 0.78 %, 1.64 %, and 1.19 % on the same datasets for SOH estimation. While the performance on the LFP dataset is comparable, our model demonstrates notable improvements on the NCA and NCM datasets. Considering the potential differences

Table 3
Performance comparison of models.

		SOH		S-DP		RUL	
		MAPE	RMSE	MAPE	RMSE	MAE	RMSE
LFP	MLP	1.37 %	0.0194	1.33 %	0.0187	162	196
	Bi-LSTM	1.16 %	0.0161	1.14 %	0.0156	143	187
	Trans	1.04 %	0.0143	1.08 %	0.0141	129	169
	This work	0.85 %	0.0121	0.79 %	0.0113	117	132
NCA	MLP	1.53 %	0.0162	1.64 %	0.0189	154	187
	Bi-LSTM	1.33 %	0.0145	1.49 %	0.0162	118	169
	Trans	1.20 %	0.0123	1.19 %	0.0126	114	164
	This work	1.08 %	0.0105	0.92 %	0.0094	104	146
NCM	MLP	1.21 %	0.0132	1.27 %	0.0136	132	190
	Bi-LSTM	1.01 %	0.0113	1.07 %	0.0114	128	178
	Trans	0.80 %	0.0086	0.90 %	0.0099	126	170
	This work	0.75 %	0.0079	0.83 %	0.0093	121	163

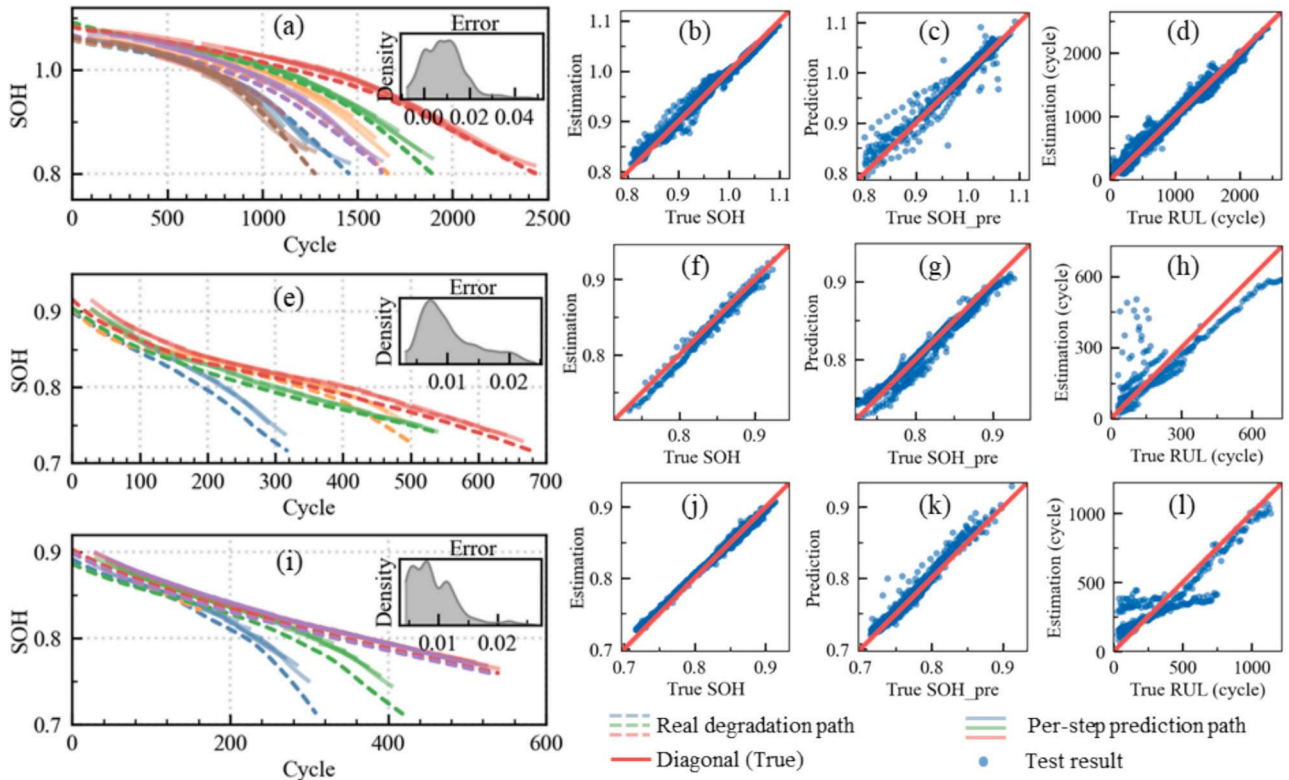


Fig. 5. Test results of the multi-task model for batteries with different chemistries. Subplots (a–d) correspond to LFP batteries, (e–h) to NCA batteries, and (i–l) to NCM batteries. Specifically: (a, e, i) show the short-term degraded path prediction results; (b, f, j) illustrate the distribution of SOH estimation results; (c, g, k) display the distributions of short-term degraded path predictions across all test batteries; and (d, h, l) depict the distribution of RUL prediction results.

between the two studies in terms of task quantity, dataset splitting, model configuration, and training hyperparameters, the slight variations observed in the LFP results are deemed acceptable. This comparison on the LFP dataset also highlights the profound impact of battery chemistry and testing protocols on model performance.

All experiments were conducted on a workstation equipped with an Intel i5-13600KF CPU, 64 GB RAM, and an NVIDIA RTX 4060 Ti GPU (CUDA 12.4.127). On the LFP dataset, the training times for MLP, Bi-LSTM, Transformer, and the proposed model were 2.1, 5.9, 4.3, and 21.2 min, respectively. To ensure consistency, all reported values are averaged over three runs. The increased training time of our model results from its greater architectural complexity and the use of a sliding-window framework for short-term degradation prediction. Nonetheless, the model maintains efficient inference performance, with latency measured at 0.069 ms on GPU and 0.114 ms on CPU—comparable to baseline models such as Bi-LSTM (0.032 ms GPU / 0.041 ms CPU). Training memory usage also remained within a reasonable range: 139.2 MB on GPU and 252.8 MB on CPU, which is slightly higher than baseline models like Transformer (91.4 MB GPU / 128.1 MB CPU), which is acceptable given the model's additional complexity. Notably, once trained, the model performs inference using only basic matrix operations, without iterative procedures or derivative computations, making it well-suited for real-time deployment.

4.2. Degraded path prediction

The degradation path of lithium-ion batteries delineates the evolution of the SOH over an extended temporal scale, offering a comprehensive characterization of the battery aging process. It not only reflects the current health status but also unveils how the battery progressively approaches its end of life under the cumulative effect of charge-discharge cycles. The prediction of degradation trajectories falls under the purview of long-term time series forecasting, typically necessitating the employment of a sliding window approach for iterative prediction. In the model we have proposed, a Bi-LSTM network is utilized to forecast the degradation path. However, as the prediction time extends, the model's predictions tend to gradually deviate from the true values. It remains difficult to accurately reconstruct the full degradation path of a battery solely from externally measurable signals such as voltage and current during charge/discharge cycles. Therefore, we focus solely on short-term predictions for the degradation path to ensure prediction accuracy and reliability.

As illustrated in Fig. 5(a), (e), and (i), representative curves of degradation path predictions are depicted, while Fig. 5(c), (g), and (k) present the diagonal plots of the prediction results. The x-axis corresponds to the predicted SOH values produced by the model over the short-term future period, while the y-axis shows the corresponding ground truth values. In different datasets, we dynamically adjust the length of S-DP predictions to accommodate their respective degradation characteristics. Specifically, the S-DP prediction length for the LFP dataset is set at 600 cycles, whereas for the NCA and NCM datasets, it is set at 150 cycles. To exhibit the distribution of prediction errors, we have plotted the error density distribution graphs. These graphs reveal that our proposed model performs well in the S-DP predictions of degradation trajectories, with errors predominantly clustered around 0.01. To further corroborate the model's superiority, we conducted comparative experiments with other models. Given that traditional models struggle with iterative prediction, we uniformly set the SOH prediction length for all models to 30 cycles. The results, as shown in Table 3, demonstrate that our model outperforms other comparative models in the S-DP prediction across all three datasets.

4.3. RUL prediction

To ensure the safe use of batteries, predicting the remaining time until the end of their life is significant. The remaining lifespan can be

divided into calendar life and cycle life. Calendar life refers to the remaining days of use while the battery is in storage, whereas cycle life refers to the remaining number of charge-discharge cycles under the variable operating conditions of mobile devices. We focus on predicting the remaining cycle life of electric vehicle batteries under different battery materials, usage temperatures, and charge-discharge strategies. Fig. 5(d), (h), and (l) display the prediction results. It can be observed that the predictions are significantly more accurate in the early stages of the battery's life compared to the later stages, indicating that our feature extraction method is more suitable for predicting the battery's early life. However, the model's RUL prediction performance is poorer in the later stages of life for the NCA and NCM datasets, primarily due to larger temperature variations and fewer samples under consistent conditions. These factors limit the model's ability to learn stable degradation patterns, rather than indicating a deficiency in model capacity. Moreover, the strongly nonlinear degradation behavior of batteries with these electrode chemistries, combined with the lack of explicit modeling of environmental influences such as temperature, further impacts RUL accuracy. This is consistent with the feature correlation analysis in Fig. 2, where weaker correlations with RUL suggest reduced feature effectiveness in capturing degradation trends.

As demonstrated by the model comparison results in Table 3, our developed model also exhibits leading performance in the RUL prediction task. The datasets used in this study exhibit clear nonlinear degradation patterns. The LFP dataset is identical to that used in [67] for nonlinear aging knee-point prediction, while the NCA and NCM datasets demonstrate accelerated degradation in later lifecycle stages and were also used in [68] for predicting nonlinear knee-points and remaining useful life. The strong performance of our model on these datasets provides preliminary evidence of its ability to capture such complex degradation behaviors. In future work, we aim to explore more robust and training-friendly physics-informed constraints, such as sign constraints, physics-guided priors, or decoupled monotonicity losses, to further enhance physical consistency under extreme aging conditions.

4.4. Sensitivity analyses of the training sample size

The scarcity and high value of battery operational data necessitate that models exhibit robust performance under small-sample training conditions. To systematically evaluate the performance of models with limited training samples, we utilize the LFP battery dataset to conduct a comparative analysis of the performance evolution of different models as the number of training samples decreases. Fig. 6 presents the experimental results of the MAPE and RMSE metrics for two critical tasks: SOH estimation and RUL prediction (Since S-DP prediction and SOH estimation exhibit similarities in model comparison, the results of S-DP prediction are not presented separately). The experimental results indicate that, as the scale of training samples diminishes, the performance of all models exhibits the expected decline. However, across different data scales, the model we proposed consistently demonstrates significant competitive advantages. Particularly in extreme small-sample training scenarios (such as when only 5 or 15 batteries of experimental data are used for model training), the model we proposed shows superior robustness in key performance metrics compared to the contrast models, with its performance advantages being more pronounced than those of traditional methods. It thoroughly validates the technical advancement of the proposed method in small-sample learning scenarios.

4.5. Transfer learning

Transfer learning, as an innovative paradigm for cross-domain knowledge transfer, effectively overcomes the bottleneck of data scarcity in the target domain. We construct a transfer learning framework based on differences in battery material, transferring prior knowledge gained from training on Group A battery (source domain) to Group B

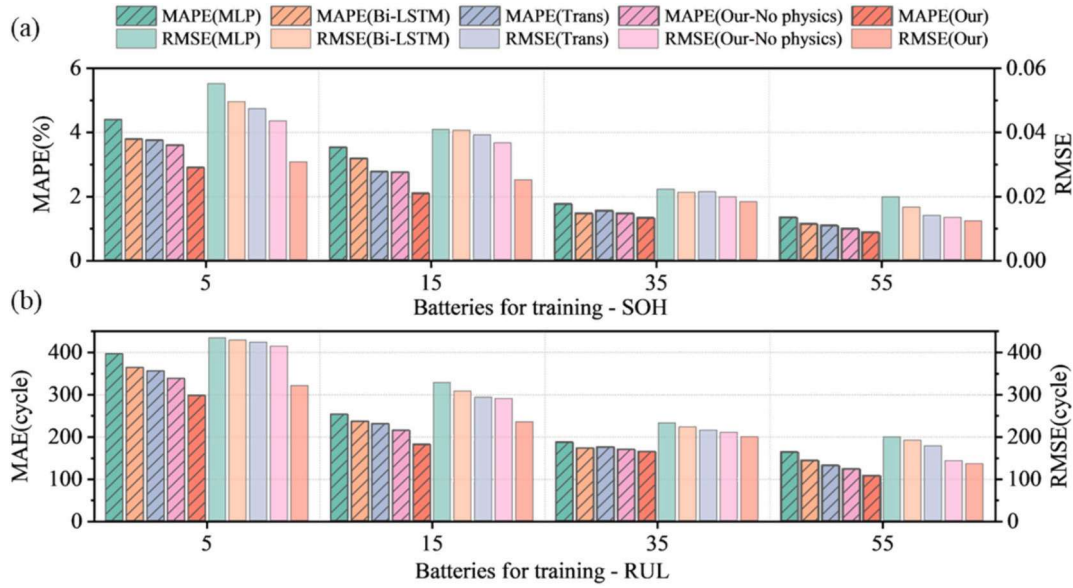


Fig. 6. Performance under different sizes of training sets.

battery (target domain). This not only significantly alleviates the constraints of insufficient data in the target domain but also enhances the model's convergence speed. As shown in Fig. 7, at the technical implementation level, we employ a parameter freezing strategy to fix the network weights of the physical feature extraction layer [48], while adaptively optimizing the remaining trainable parameters of the model with a fine-tuning learning rate on the order of $1e-4$. Table 4 systematically presents the evaluation results of the proposed transfer learning framework across battery datasets with different materials. Taking the first row as an example: the model is pre-trained using LFP battery data as the source domain, followed by parameter fine-tuning on target domain datasets such as NCA and NCM (each target domain uses only 2 batteries of data). The bolded values represent the performance of the baseline model without transfer learning (i.e., relying entirely on 2 batteries of data from the target domain for training). The experimental data clearly shows that after introducing the transfer learning mechanism, the model's errors in different tasks are significantly reduced, strongly validating the effectiveness of the transfer strategy based on physical information sharing in cross-material battery health

Table 4

Results of Transfer Learning, evaluated using RMSE.

		LFP_traget	NCA_traget	NCM_traget
SOH	LFP	0.0557	0.0346	0.0266
	NCA	0.0495	0.0371	0.0251
	NCM	0.0508	0.0315	0.0319
RUL	LFP	440	215	190
	NCA	398	251	184
	NCM	421	202	207
S-DP	LFP	0.0706	0.0369	0.0403
	NCA	0.0517	0.0408	0.0395
	NCM	0.0553	0.0357	0.0425

management.

5. Conclusion

To address the issue of battery health management, we propose a novel approach that integrates a multi-task processing framework with a physics-informed network. The core of this approach lies in leveraging shared and independent parameters to achieve task synergy while utilizing PINN to uncover the underlying degradation physics within the data. Specifically, the method extracts features from voltage and current data segments around the constant voltage charging phase, providing broad applicability to battery data across diverse materials and operating conditions. Additionally, a simplified Transformer model is employed to extract temporal and feature information from historical data, and an adaptive weighting method is used to synthesize the loss of various tasks. Experimental results demonstrate that the model achieves MAPE of 0.75 % in SOH estimation, an average deviation of approximately 0.01 in S-DP prediction, and MAE of 104 cycles in RUL prediction, all of which perform excellently in comparative experiments. Furthermore, we conduct an in-depth analysis of the impact of training sample size on model performance and explore transfer learning. In summary, the proposed multi-task health management method not only enables reliable and accurate task predictions but also offers advantages in terms of wide applicability and strong generalization capability.

The proposed method demonstrates strong potential in applications such as electric vehicles and energy storage systems, but several challenges remain for industrial deployment. In real-world scenarios, low-temperature environments often introduce distinct battery degradation

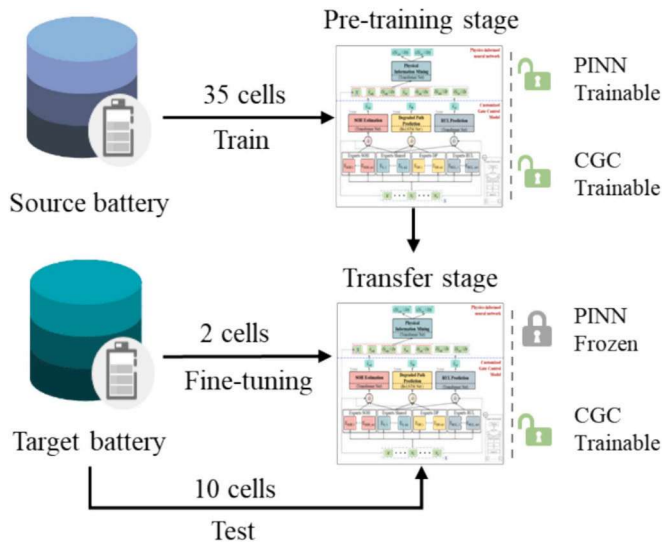


Fig. 7. Schematic Diagram of Transfer Learning.

mechanisms. Evaluating the model under such conditions is crucial for assessing its robustness and adaptability. To support deployment efficiency, combining cloud-based training with local inference can help balance computational cost and responsiveness, especially when paired with automatic hyperparameter tuning strategies. Additionally, the current model is based on empirical formulas with limited physical interpretability. Integrating micro-level electrochemical aging mechanisms would improve physical consistency and enhance trustworthiness in engineering applications.

CRedit authorship contribution statement

Li Yang: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Mingjian He:** Supervision, Project administration. **Yatao Ren:** Writing – review & editing, Resources. **Baohai Gao:** Writing – review & editing, Supervision, Data curation. **Hong Qi:** Project administration, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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